Technische Universität München
<TUM School/Fakultät/ TUM Campus Straubing>

# <Titel der Dissertation>

Vorname(n) Name

Vollständiger Abdruck der von der/des <TUM School of/Fakultät für/ TUM Campus Straubing> der Technischen Universität München zur Erlangung eines/r

Doktors/Doktorin der …… (Dr. ….)

genehmigten Dissertation.

Vorsitz: ……

Prüfer*innen der Dissertation:

1. ……
2. ……
3. ……

Die Dissertation wurde am ..... bei der Technischen Universität München eingereicht und durch die/den <TUM School/Fakultät/ TUM Campus Straubing> am ..... angenommen

## Acknowledgements

These are the acknowledgements. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

## Abstract

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# 1

# Introduction

Note that you may have multiple `\include` statements here, e.g. one for each subsection.

General structure of this chapter should read as follows. This chapter should be used to motivate your study and answer the question "Why is this important?". Also, it should define what you set out to achieve (these will be revisited in the conclusions chapter). You should describe your approach to the Aims and Objectives, including an evaluation part.

## 1.1 | Motivation

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

## 1.2 | Aims and Objectives

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

## 1.3 | Our Approach

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

## 1.4 | Contributions

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

## 1.5 | Document Structure

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

# 2

# Background & Literature Overview

In this section you need to explain all the theory required to understand your dissertation (i.e. the following chapters). But really in this chapter I am going to show you some examples.

## 2.1 | Some Technique One

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

### 2.1.1 | Some Sub-technique One

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This

text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

#### 2.1.1.1 | Some Sub-sub-technique One

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

## 2.2 | Some Technique Two with Super Long Title Which Will Overrun In Header

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Imagine some colourful description on Some Technique Three.

## 2.3 | Evaluation Criteria

This section should contain information on the metrics and background used to evaluate your work.

## 2.4 | Related Work

**In this section you need to explain (and reference) similar work in literature**. Make sure to:

- Give a systematic overview of papers with related/similar work
- Highlight similarities/differences to your work (perhaps in the form of a table)

Note that this section may be sectioned based on the different aspects of your dissertation.

## 2.5 | Summary

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

# 3

# Materials & Methods

This section should include a recipe of what you did (explain what you have done so if someone wants to reproduce the experiment, they can). A flow chart is typically helpful. Also, make sure to define all software that you used including version numbers and OS. Should also include a description of statistical methods used (if any).[1]

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

## 3.1 | Subsections and Subsubsections

This is a section.

### 3.1.1 | A Subsection

This is a subsection.

#### 3.1.1.1 | A Subsubsection

This is a subsubsection.

[1] For more information see: http://rc.rcjournal.com/content/49/10/1229.short. Last Accessed: 27th November, 2024.

## 3.2 | Footnotes

Some text with a footnote, if an online link remember to add *Last Accessed*.[2]

## 3.3 | Equations

The following is the most beautiful equation in maths, Euler's Identity (Equation 3.1).

$$e^{i\pi} + 1 = 0 \tag{3.1}$$

where:

$e$ = the constant
$i$ = of complex fame
$\pi$ = not of the apple variety

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

## 3.4 | Numbered Lists

This is an example of a numbered list:

1. This is my first point
2. My second
3. My third!
4. And my fourth?

[2] Some footnote text.

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

## 3.5 | Bulleted Lists

This is an example of a bulleted list:

- This is my first point
- My second
- My third!
- And my fourth?

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

## 3.6 | Figures

A test figure is shown in Figure 3.1.

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

**Figure 3.1:** A test figure. This caption is huge, but in the list of figures only the smaller version in the square brackets will appear.

## 3.7 | Two Side-by-Side Figures

Two figures shown side-by-side are shown in Figure 3.2.

(a) Goku

(b) More Goku

**Figure 3.2:** The same super saiyan. Two times.

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no in-

**Table 3.1:** A Beautiful and Complex Table (for tables captions above)

| | $w = 8$ | | | $w = 16$ | | |
|---|---|---|---|---|---|---|
| | $t = 0$ | $t = 1$ | $t = 2$ | $t = 0$ | $t = 1$ | $t = 2$ |
| $dir = 1$ | | | | | | |
| $c$ | 0.0790 | 0.1692 | 0.2945 | 0.3670 | 0.7187 | 3.1815 |
| $c$ | -0.8651 | 50.0476 | 5.9384 | -9.0714 | 297.0923 | 46.2143 |
| $c$ | 124.2756 | -50.9612 | -14.2721 | 128.2265 | -630.5455 | -381.0930 |
| $dir = 0$ | | | | | | |
| $c$ | 0.0357 | 1.2473 | 0.2119 | 0.3593 | -0.2755 | 2.1764 |
| $c$ | -17.9048 | -37.1111 | 8.8591 | -30.7381 | -9.5952 | -3.0000 |
| $c$ | 105.5518 | 232.1160 | -94.7351 | 100.2497 | 141.2778 | -259.7326 |

formation? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

## 3.8 | Acronyms

In the early nineties, **GSM!** was deployed in many European countries. **GSM!** (**GSM!**) offered for the first time international roaming for mobile subscribers. The **GSM!**'s use of **TDMA!** (**TDMA!**) as its communication standard was debated at length. And every now and then there are big discussion whether **CDMA!** (**CDMA!**) should have been chosen over **TDMA!**.

If you want to know more about **GSM!** (**GSM!**), **TDMA!** (**TDMA!**), **CDMA!** (**CDMA!**) and other acronyms, just read a book about mobile communication. Just to mention it: There is another **UA!** (**UA!**), for testing.

## 3.9 | Tables

A beautiful table is shown in Table 3.1, data from Ebejer et al. (2012).

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no in-

formation? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

## 3.10 | Long Tables

The following is an example of a table (Table 3.2) spanning multiple pages.

**Table 3.2:** Performance of Ligity in HTS mode against the Ligity-compatible DUD-E targets. The mean (and standard deviation in parentheses) values of ROC AUC using Tanimoto is 0.622 ($\pm 0.132$), while for Tversky it is 0.671 ($\pm 0.142$); the mean $EF_{1\%}$ using Tanimoto is 5.648 ($\pm 8.668$), while for $EF_{1\%}$ using Tversky it is 9.047 ($\pm 12.713$).

| Target | No. of Actives | No. of Decoys | ROC AUC Tanimoto | ROC AUC Tversky | BEDROC Tanimoto | BEDROC Tversky | $EF_{1\%}$ Tanimoto | $EF_{1\%}$ Tversky |
|---|---|---|---|---|---|---|---|---|
| ABL1 | 182 | 10,750 | 0.563 | 0.473 | 0.077 | 0.077 | 1.653 | 2.204 |
| ACE | 281 | 16,877 | 0.787 | 0.787 | 0.336 | 0.401 | 12.425 | 19.525 |
| ACES | 453 | 26,242 | 0.634 | 0.645 | 0.077 | 0.155 | 1.766 | 5.518 |
| ADA | 93 | 5,450 | 0.724 | 0.660 | 0.149 | 0.147 | 3.251 | 3.251 |
| ADA17 | 532 | 35,898 | 0.638 | 0.728 | 0.103 | 0.283 | 1.317 | 9.030 |
| ADRB1 | 247 | 15,850 | 0.523 | 0.647 | 0.065 | 0.129 | 1.619 | 5.262 |
| ADRB2 | 231 | 14,999 | 0.523 | 0.589 | 0.052 | 0.040 | 1.735 | 0.000 |
| AKT1 | 293 | 16,450 | 0.386 | 0.548 | 0.039 | 0.107 | 2.737 | 3.080 |
| AKT2 | 117 | 6,900 | 0.511 | 0.685 | 0.140 | 0.194 | 8.568 | 8.568 |
| ALDR | 159 | 8,988 | 0.574 | 0.610 | 0.202 | 0.172 | 10.747 | 6.322 |
| AMPC | 48 | 2,845 | 0.521 | 0.541 | 0.049 | 0.023 | 0.000 | 0.000 |
| ANDR | 269 | 14,349 | 0.722 | 0.742 | 0.194 | 0.354 | 4.839 | 24.938 |
| AOFB | 121 | 6,875 | 0.422 | 0.464 | 0.045 | 0.027 | 1.652 | 0.000 |
| BACE1 | 283 | 18,100 | 0.441 | 0.775 | 0.017 | 0.310 | 0.000 | 13.062 |
| BRAF | 152 | 9,950 | 0.612 | 0.639 | 0.208 | 0.165 | 12.502 | 5.264 |
| CASP3 | 199 | 10,694 | 0.600 | 0.734 | 0.068 | 0.258 | 0.502 | 7.031 |
| CDK2 | 474 | 27,838 | 0.467 | 0.507 | 0.021 | 0.048 | 0.000 | 1.055 |
| COMT | 41 | 3,846 | 0.789 | 0.889 | 0.338 | 0.665 | 19.447 | 58.341 |
| CP2C9 | 120 | 7,449 | 0.518 | 0.634 | 0.058 | 0.186 | 1.660 | 8.299 |
| CP3A4 | 170 | 11,787 | 0.450 | 0.493 | 0.022 | 0.057 | 0.000 | 2.345 |
| CSF1R | 166 | 12,149 | 0.526 | 0.542 | 0.136 | 0.152 | 6.031 | 7.238 |
| CXCR4 | 40 | 3,405 | 0.575 | 0.722 | 0.217 | 0.134 | 12.665 | 0.000 |
| DEF | 102 | 5,699 | 0.732 | 0.833 | 0.212 | 0.379 | 10.786 | 15.689 |

(continued...)

| Target | No. of Actives | No. of Decoys | ROC AUC Tanimoto | ROC AUC Tversky | BEDROC Tanimoto | BEDROC Tversky | $EF_{1\%}$ Tanimoto | $EF_{1\%}$ Tversky |
|---|---|---|---|---|---|---|---|---|
| DHI1 | 330 | 19,348 | 0.481 | 0.595 | 0.089 | 0.062 | 2.422 | 1.211 |
| DPP4 | 533 | 40,941 | 0.586 | 0.591 | 0.154 | 0.157 | 4.312 | 3.937 |
| DRD3 | 480 | 34,048 | 0.484 | 0.441 | 0.043 | 0.046 | 1.251 | 0.626 |
| DYR | 231 | 17,196 | 0.694 | 0.758 | 0.210 | 0.230 | 6.504 | 7.371 |
| EGFR | 542 | 35,047 | 0.593 | 0.491 | 0.054 | 0.037 | 0.922 | 0.000 |
| ESR1 | 383 | 20,683 | 0.838 | 0.861 | 0.527 | 0.594 | 31.281 | 39.101 |
| ESR2 | 367 | 20,199 | 0.844 | 0.870 | 0.563 | 0.644 | 20.130 | 32.644 |
| FA10 | 537 | 28,324 | 0.564 | 0.674 | 0.058 | 0.118 | 0.930 | 2.232 |
| FA7 | 114 | 6,249 | 0.762 | 0.859 | 0.210 | 0.332 | 6.105 | 8.721 |
| FABP4 | 47 | 2,749 | 0.786 | 0.744 | 0.191 | 0.276 | 0.000 | 10.623 |
| FAK1 | 100 | 5,350 | 0.642 | 0.531 | 0.111 | 0.065 | 2.019 | 0.000 |
| FGFR1 | 139 | 8,698 | 0.511 | 0.522 | 0.036 | 0.088 | 0.722 | 1.445 |
| FKB1A | 111 | 5,799 | 0.605 | 0.751 | 0.162 | 0.164 | 8.122 | 3.610 |
| FNTA | 592 | 51,493 | 0.411 | 0.625 | 0.012 | 0.132 | 0.000 | 4.053 |
| FPPS | 85 | 8,842 | 0.917 | 0.985 | 0.323 | 0.776 | 2.360 | 36.581 |
| GCR | 258 | 14,998 | 0.805 | 0.834 | 0.244 | 0.324 | 3.092 | 8.116 |
| GLCM | 54 | 3,790 | 0.667 | 0.685 | 0.182 | 0.279 | 1.873 | 11.240 |
| GRIA2 | 158 | 11,842 | 0.662 | 0.684 | 0.248 | 0.154 | 11.392 | 5.696 |
| GRIK1 | 101 | 6,547 | 0.656 | 0.668 | 0.203 | 0.102 | 7.978 | 1.995 |
| HDAC2 | 185 | 10,300 | 0.676 | 0.734 | 0.187 | 0.201 | 4.318 | 4.318 |
| HDAC8 | 170 | 10,449 | 0.640 | 0.819 | 0.120 | 0.377 | 2.946 | 8.250 |
| HIVINT | 100 | 6,640 | 0.390 | 0.554 | 0.030 | 0.116 | 0.000 | 3.018 |
| HIVPR | 535 | 35,724 | 0.663 | 0.872 | 0.072 | 0.490 | 0.187 | 23.898 |
| HIVRT | 338 | 18,884 | 0.495 | 0.475 | 0.124 | 0.085 | 4.443 | 1.777 |
| HMDH | 170 | 8,750 | 0.480 | 0.906 | 0.068 | 0.652 | 2.358 | 35.963 |
| HS90A | 88 | 4,850 | 0.635 | 0.506 | 0.096 | 0.083 | 0.000 | 3.436 |
| HXK4 | 92 | 4,700 | 0.662 | 0.803 | 0.206 | 0.307 | 15.192 | 9.766 |
| IGF1R | 148 | 9,300 | 0.502 | 0.575 | 0.057 | 0.189 | 2.037 | 14.941 |
| INHA | 43 | 2,300 | 0.493 | 0.575 | 0.031 | 0.045 | 0.000 | 0.000 |
| ITAL | 138 | 8,500 | 0.619 | 0.465 | 0.037 | 0.065 | 0.000 | 0.728 |
| JAK2 | 107 | 6,500 | 0.472 | 0.475 | 0.073 | 0.118 | 2.807 | 6.549 |
| KIF11 | 116 | 6,850 | 0.755 | 0.781 | 0.149 | 0.219 | 4.289 | 2.574 |
| KIT | 166 | 10,449 | 0.463 | 0.437 | 0.045 | 0.030 | 0.000 | 0.000 |
| KITH | 57 | 2,850 | 0.649 | 0.838 | 0.228 | 0.709 | 14.069 | 47.483 |
| KPCB | 135 | 8,699 | 0.753 | 0.813 | 0.220 | 0.338 | 8.923 | 12.641 |
| LCK | 419 | 27,391 | 0.471 | 0.437 | 0.031 | 0.043 | 0.000 | 1.910 |
| LKHA4 | 171 | 9,448 | 0.718 | 0.694 | 0.238 | 0.150 | 8.203 | 1.758 |
| MAPK2 | 101 | 6,148 | 0.660 | 0.670 | 0.174 | 0.199 | 5.988 | 3.992 |
| MCR | 94 | 5,149 | 0.816 | 0.888 | 0.215 | 0.454 | 6.436 | 19.307 |
| MET | 166 | 11,249 | 0.566 | 0.531 | 0.130 | 0.065 | 6.032 | 0.603 |
| MK01 | 79 | 4,550 | 0.518 | 0.602 | 0.121 | 0.206 | 5.095 | 3.821 |
| MK10 | 104 | 6,600 | 0.488 | 0.489 | 0.020 | 0.031 | 0.962 | 0.962 |
| MK14 | 578 | 35,847 | 0.511 | 0.589 | 0.040 | 0.064 | 0.173 | 0.519 |
| MMP13 | 572 | 37,199 | 0.648 | 0.753 | 0.134 | 0.268 | 2.446 | 9.957 |
| MP2K1 | 121 | 8,146 | 0.669 | 0.569 | 0.187 | 0.058 | 3.293 | 0.823 |
| NOS1 | 98 | 8,028 | 0.483 | 0.451 | 0.109 | 0.041 | 3.071 | 0.000 |

(continued...)

| Target | No. of Actives | No. of Decoys | ROC AUC Tanimoto | ROC AUC Tversky | BEDROC Tanimoto | BEDROC Tversky | $EF_{1\%}$ Tanimoto | $EF_{1\%}$ Tversky |
|---|---|---|---|---|---|---|---|---|
| NRAM | 98 | 6,200 | 0.853 | 0.859 | 0.342 | 0.290 | 11.221 | 3.060 |
| PA2GA | 99 | 5,150 | 0.793 | 0.756 | 0.225 | 0.153 | 1.020 | 3.059 |
| PARP1 | 508 | 30,029 | 0.635 | 0.692 | 0.215 | 0.231 | 11.234 | 7.884 |
| PGH1 | 195 | 10,798 | 0.645 | 0.637 | 0.077 | 0.100 | 0.000 | 2.050 |
| PGH2 | 435 | 23,139 | 0.716 | 0.780 | 0.166 | 0.291 | 3.444 | 9.874 |
| PLK1 | 107 | 6,800 | 0.658 | 0.531 | 0.123 | 0.048 | 1.871 | 0.000 |
| PNPH | 103 | 6,946 | 0.575 | 0.578 | 0.161 | 0.181 | 4.888 | 8.799 |
| PPARA | 373 | 19,399 | 0.783 | 0.778 | 0.262 | 0.280 | 6.693 | 7.764 |
| PPARD | 240 | 12,250 | 0.547 | 0.544 | 0.078 | 0.098 | 1.665 | 2.498 |
| PPARG | 484 | 25,299 | 0.515 | 0.605 | 0.055 | 0.118 | 0.619 | 4.955 |
| PRGR | 293 | 15,648 | 0.740 | 0.793 | 0.142 | 0.318 | 2.053 | 14.714 |
| PTN1 | 130 | 7,249 | 0.398 | 0.538 | 0.055 | 0.090 | 0.000 | 3.068 |
| PUR2 | 50 | 2,700 | 0.851 | 0.837 | 0.281 | 0.255 | 7.857 | 1.964 |
| PYGM | 77 | 3,944 | 0.403 | 0.492 | 0.016 | 0.137 | 0.000 | 3.917 |
| PYRD | 111 | 6,449 | 0.682 | 0.710 | 0.462 | 0.413 | 34.027 | 16.118 |
| RENI | 104 | 6,956 | 0.720 | 0.789 | 0.043 | 0.138 | 0.000 | 0.000 |
| ROCK1 | 100 | 6,300 | 0.347 | 0.449 | 0.020 | 0.084 | 1.000 | 4.000 |
| RXRA | 131 | 6,950 | 0.788 | 0.900 | 0.219 | 0.596 | 6.091 | 27.407 |
| SAHH | 63 | 3,450 | 0.874 | 0.852 | 0.598 | 0.542 | 35.050 | 27.084 |
| SRC | 524 | 34,500 | 0.565 | 0.477 | 0.065 | 0.050 | 0.382 | 0.573 |
| TGFR1 | 133 | 8,499 | 0.609 | 0.639 | 0.147 | 0.154 | 10.565 | 4.528 |
| THB | 103 | 7,450 | 0.794 | 0.762 | 0.238 | 0.150 | 10.614 | 0.965 |
| THRB | 461 | 27,000 | 0.605 | 0.706 | 0.063 | 0.166 | 2.166 | 5.632 |
| TRY1 | 449 | 25,975 | 0.711 | 0.815 | 0.147 | 0.280 | 2.898 | 6.688 |
| TRYB1 | 148 | 7,650 | 0.670 | 0.670 | 0.153 | 0.132 | 3.378 | 3.378 |
| TYSY | 109 | 6,745 | 0.594 | 0.725 | 0.071 | 0.226 | 0.911 | 5.468 |
| UROK | 162 | 9,850 | 0.525 | 0.650 | 0.036 | 0.120 | 0.000 | 1.854 |
| VGFR2 | 409 | 24,948 | 0.632 | 0.578 | 0.083 | 0.093 | 1.465 | 1.465 |
| WEE1 | 102 | 6,150 | 0.934 | 0.929 | 0.789 | 0.797 | 59.348 | 61.294 |
| XIAP | 100 | 5,150 | 0.752 | 0.974 | 0.190 | 0.897 | 8.077 | 51.490 |

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## 3.11 | Landscape Tables

Next is an example of a wide table on a landscape oriented paper (Table 3.3).

**Table 3.3:** A table in landscape orientation.

| $m$ | $x$ | $y$ | $z$ | $a$ | $A_m$ | $B$ | $C$ | $x$ | $y$ | $z$ | $a$ | $A_m$ | $B$ | $C$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 16.128 | +8.872 | 16.128 | 1.402 | 1.373 | -146.6 | -137.6 | 16.128 | +8.872 | 16.128 | 1.402 | 1.373 | -146.6 | -137.6 |
| 2 | 3.442 | -2.509 | 3.442 | 0.299 | 0.343 | 133.2 | 152.4 | 3.442 | -2.509 | 3.442 | 0.299 | 0.343 | 133.2 | 152.4 |
| 3 | 1.826 | -0.363 | 1.826 | 0.159 | 0.119 | 168.5 | -161.1 | 1.826 | -0.363 | 1.826 | 0.159 | 0.119 | 168.5 | -161.1 |
| 4 | 0.993 | -0.429 | 0.993 | 0.086 | 0.08 | 25.6 | 90 | 1.826 | -0.363 | 1.826 | 0.159 | 0.119 | 168.5 | -161.1 |
| 5 | 1.29 | +0.099 | 1.29 | 0.112 | 0.097 | -175.6 | -114.7 | 1.826 | -0.363 | 1.826 | 0.159 | 0.119 | 168.5 | -161.1 |
| 6 | 0.483 | -0.183 | 0.483 | 0.042 | 0.063 | 22.3 | 122.5 | 1.826 | -0.363 | 1.826 | 0.159 | 0.119 | 168.5 | -161.1 |
| 7 | 0.766 | -0.475 | 0.766 | 0.067 | 0.039 | 141.6 | -122 | 1.826 | -0.363 | 1.826 | 0.159 | 0.119 | 168.5 | -161.1 |
| 8 | 0.624 | +0.365 | 0.624 | 0.054 | 0.04 | -35.7 | 90 | 1.826 | -0.363 | 1.826 | 0.159 | 0.119 | 168.5 | -161.1 |
| 9 | 0.641 | -0.466 | 0.641 | 0.056 | 0.045 | 133.3 | -106.3 | 1.826 | -0.363 | 1.826 | 0.159 | 0.119 | 168.5 | -161.1 |
| 10 | 0.45 | +0.421 | 0.45 | 0.039 | 0.034 | -69.4 | 110.9 | 1.826 | -0.363 | 1.826 | 0.159 | 0.119 | 168.5 | -161.1 |
| 11 | 0.598 | -0.597 | 0.598 | 0.052 | 0.025 | 92.3 | -109.3 | 1.826 | -0.363 | 1.826 | 0.159 | 0.119 | 168.5 | -161.1 |

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## 3.12 | Theorems

**Theorem 3.12.1.** *Let $f$ be a function whose derivative exists in every point, then $f$ is a continuous function.*

**Theorem 3.12.2** (Pythagorean theorem)**.** *This is a theorem about right triangles and can be summarised in the next equation*

$$x^2 + y^2 = z^2$$

And a consequence of Theorem 3.12.2 is the statement in the next corollary.

**Corollary 3.12.2.1.** *There's no right rectangle whose sides measure 3 cm, 4 cm, and 6 cm.*

You can reference theorems such as 3.12.2 when a label is assigned.

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## 3.13 | Lemmas

**Lemma 3.13.1.** *Given two line segments whose lengths are $a$ and $b$ respectively there is a real number $r$ such that $b = ra$.*

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Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

## 3.14 | Proofs

**Lemma 3.14.1.** *Given two line segments whose lengths are $a$ and $b$ respectively there is a real number $r$ such that $b = ra$.*

*Proof.* To prove it by contradiction try and assume that the statement is false, proceed from there and at some point you will arrive to a contradiction. □

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

## 3.15 | Code Listings

Here you go.

```
import numpy as np

def incmatrix(genl1,genl2):
m = len(genl1)
n = len(genl2)
M = None #to become the incidence matrix
VT = np.zeros((n*m,1), int)  #dummy variable

#compute the bitwise xor matrix
M1 = bitxormatrix(genl1)
M2 = np.triu(bitxormatrix(genl2),1)

for i in range(m-1):
for j in range(i+1, m):
[r,c] = np.where(M2 == M1[i,j])
```

```
for k in range(len(r)):
VT[(i)*n + r[k]] = 1;
VT[(i)*n + c[k]] = 1;
VT[(j)*n + r[k]] = 1;
VT[(j)*n + c[k]] = 1;

if M is None:
M = np.copy(VT)
else:
M = np.concatenate((M, VT), 1)

VT = np.zeros((n*m,1), int)

return M
```

**Listing 3.1:** My Listing Caption

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

## 3.16 | Algorithms

An algorithm example is shown in Algorithm 1. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

## 3.17 | Suppressing Page Numbers on a Float Page

Kindly refer to Figure 3.3.

**Algorithm 1:** An algorithm with caption

```
Data: n ≥ 0
Result: y = x^n
y ← 1;
X ← x;
N ← n;
while N ≠ 0 do
    if N is even then
        X ← X × X;
        N ← N/2 ;                    /* This is a comment */
    else
        if N is odd then
            y ← y × X;
            N ← N − 1;
        end
    end
end
```

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

**Figure 3.3:** Page numbers are suppressed on this page.

## 3.18 | Referencing

Use `\textcite` for in-text citations, e.g. Einstein (1905), and `\parencite` for citations in parenthesis. And this is what an online reference looks like (LLC, 1999).

In their study, Einstein (1905) show the world is round. Others have shown this to be the case (Arrighi, 2003; Ebejer et al., 2016).

## 3.19 | Using a Different Language

Local name = Malti. Today = **?today?**. Jiena ngħix Ħaż-Żebbuġ.

## 3.20 | Summary

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

# 4

# Results & Discussion

Should include a reiteration of the experiments, and their outcome. Together with a description (discussion). Preamble should include a reminder of the aims and objectives together with a list of experiments to achieve these. Should include many charts and other visualization with appropriate descriptions.[1]

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

## 4.1 | Summary

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

[1] Another footnote example.

# 5

# Evaluation

In an ideal world, you should have two kind of evaluations. The first is against some ground truth (perhaps a random model?). The second kind of evaluation is against other people's work (accuracy, speed, etc.). Any dimension which is of interest, should be evaluated. Evaluation should be statistically sound.

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

## 5.1 | Summary

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# 6

# Conclusion

This section should have a summary of the whole project. The original aims and objective and whether these have been met should be discussed. It should include a section with a critique and a list of limitations of your proposed solutions. Future work should be described, and this should not be marginal or silly (e.g. add machine learning models). It is always good to end on a positive note (i.e. 'Final Remarks').

## 6.1 | Revisiting the Aims and Objectives

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

## 6.2 | Critique and Limitations

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## 6.3 | Future Work

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## 6.4 | Final Remarks

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# A

# Media Content

If the dissertation has a DVD or pendrive attached to it, you will need a section which explains what is on the media (structure, files, data, etc.). This could be a table with filename and description.

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# B

# Installation Instructions

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# C

# User Manual

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# References


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